

Lierda NT26-FCN C Series (Wide Voltage Version) Hardware Design Manual

Version: Rev2.1

Date: 26/05/08

Status: Released

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Document Revision History

Document Version	Change date	Reviser	Reviewer	Change content
Rev2.0	26-01-13	HKJ		Temporary version
Rev1.0	23-12-07	CZT, HKJ	SLY, YMX	Initial version
Rev1.1	24-01-29	CZT, HKJ	SLY, YMX	Modify and expand the operating voltage range
Rev1.2	24-03-12	CWY	YHB	Optimize module dimension drawings and packaging diagrams.
Rev1.3	24-10-09	HKJ	YHB	Modify the description related to the power domain.
Rev1.4	24-10-28	CZT	SLY	Add precautions: when using the module for production, avoid environments containing mercury or mercury vapor.
Rev1.5	25-03-25	CZT		Add power outage precautions; Add section 1.1 Special Symbols; Modify the ADC sampling range; Modify the B41 working frequency band range.
Rev1.6	25-04-14	HKJ		Modify the description of the electrical section; Modify ADC sampling range
Rev1.7	25-04-17	HKJ		Add the NT26FCNC60WNA model selection, distinguishing the power consumption descriptions of the two models.
Rev1.8	25-08-08	HKJ		Add measured current and descriptions for different power consumption modes.
Rev2.0	26-03-04	HKJ		Add a description for the NT26F6C0 module; Update to the new template;
Rev2.1	26-05-08	HKJ		Modify a description of the default voltage of VDD_EXT;

Safety Instructions

Users are responsible for complying with the relevant regulations of other countries regarding wireless communication modules and devices, as well as specific usage environment laws. By following the safety principles below, personal safety can be ensured and the product and work environment can be protected from potential damage. Our company does not bear any related losses caused by customers' failure to comply with these regulations.



Road safety comes first! When you are driving, do not use handheld mobile terminal devices unless they have hands-free functionality. Please stop your vehicle before making a call!



Please turn off mobile terminal devices before boarding. The wireless functions of mobile terminals are prohibited from being turned on during the flight to prevent interference with the aircraft's communication systems. Ignoring this instruction may compromise flight safety and even violate the law.



When in hospitals or healthcare facilities, pay attention to any restrictions on the use of mobile terminal devices. RF interference can cause medical equipment to malfunction, so mobile terminal devices may need to be turned off.



Mobile terminal devices do not guarantee effective connection in all situations, such as when the mobile terminal device has no credit or the SIM is invalid. In case of emergency situations as mentioned above, please remember to use the emergency call, while ensuring that your device is powered on and within an area with sufficient signal strength.



Your mobile terminal device will receive and transmit radio frequency signals when powered on, which can cause radio frequency interference when near televisions, radios, computers, or other electronic devices.



Please keep mobile terminal devices away from flammable gases. When you are near gas stations, oil depots, chemical plants, or explosive operation sites, please turn off mobile terminal devices. Operating electronic devices in any location with potential explosion hazards poses safety risks.

Applicable Module Selection

Serial Number	Module model number	Supported frequency bands	Explanation
1	NT26F6C0	Band1/3/5/8/34/38/39/40/41*	VDD_EXT Sleep Mode Power Down Supports Wi-Fi Scan

Precautions

LTE-TDD B41 only supports 140MHz (2535-2675MHz).

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1 Introduction

This document defines the application specifications for the Lierda NT26-FCN C series modules, describing their frequency band functions, key features, hardware interfaces, operating modes, electrical characteristics, mechanical specifications, and packaging and storage. It helps users quickly grasp the application methods of this Cat.1 module for fast and flexible product design.

1.1 Special characters

Table 1-1 Explanation of Special Symbols

Symbol	Definition
*	Unless otherwise specified, an asterisk (*) marked after module functions, features, interfaces, pin names, AT commands, or parameters indicates that the module's corresponding function, feature, interface, pin name, AT command, or parameter is under development and not currently supported.

2 Product Overview

The NT26-FCN C series modules are globally leading Cat.1 wireless communication modules, compliant with the 3GPP R14 Cat.1bis standard, supporting bandwidths of 1.4/3/5/10/15/20MHz. They feature a compact size, low power consumption, and strong anti-interference capabilities. The NT26-FCN C series modules are suitable for various common IoT application scenarios, such as:

- ◆ Intelligent Meter Reading
- ◆ Intelligent Parking
- ◆ Smart City
- ◆ Intelligent Security
- ◆ Asset Tracking
- ◆ Smart Home Appliances
- ◆ Agriculture and environmental monitoring, etc.

2.1 Frequency Bands and Functions

The wireless network functions supported by the module are as follows:

Table 2-1 Module Support for Wireless Network Functions

Network functionality	Explanation
LTE-FDD	B1/B3/B5/B8
LTE-TDD	B34/B38/B39/B40/B41(140 MHz)
WIFI*	WIFI Scan

Precautions

LTE-TDD B41 only supports 140M (2535-2675MHz).

2.2 Key Features

The table below provides a description of the key product features:

Table 2-2 Description of Key Module Characteristics

Parameters	Explanation
Module encapsulation	LCC
Module size	17.7×15.8×2.4mm (L×W×H)
Module Weight	Approximately 1.5g
Operating voltage	◆ Normal operating voltage range: 3.1~4.5V, typical value: 3.6V ◆ Extended operating voltage range (1): 2.3~4.5V
VDD_EXT feature	Sleep mode: VDD_EXT power down
Ultra-low power consumption	Sleep mode: 3 μA
Transmit Power	23dBm±2.7dB(Max)
Antenna interface	50Ω characteristic impedance
USIM interface	Supported USIM card types: Class B (3.0V) and Class C (1.8V), see section 4.3 for details.
UART	1-way main serial port: ◆ Used for AT command transmission and data transfer, the supported baud rate defaults to 9600bps. ◆ The maximum baud rate supported under LPUART is 9600bps. ◆ Used for firmware upgrade, the supported baud rate defaults to 921600bps.
	1-way debugging serial port: ◆ Used for software debugging ◆ Supported default baud rate is 3Mbps
	1-way auxiliary serial port: ◆ User-defined functions
SPI*	◆ 1-channel SPI interface ◆ Compliant with SPI bus protocol specifications
I2C*	◆ 1-channel I2C interface ◆ Compliant with I2C bus protocol specifications

	◆ Supports standard mode 100KHz, fast mode 400KHz
Communication Interface Characteristics	Supports 3GPP Rel.13/14 Cat.1 radio communication interface and protocols
Network Protocol Features (2)	Supports protocols such as TCP/UDP/HTTP(S)/SSL/MQTT(S)/FOTA/PPP/RNDIS/FTP/ECM, etc.
Data Transmission Characteristics	◆ Supports up to Cat.1 FDD and TDD ◆ 1.4/3/5/10/15/20MHz RF bandwidth ◆ Uplink QPSK, 16QAM ◆ Downlink QPSK, 16QAM, 64QAM ◆ LTE-FDD: Maximum downlink speed 10Mbps, maximum uplink speed 5Mbps ◆ LTE-TDD: Maximum downlink rate 8.96 Mbps, maximum uplink rate 3.1 Mbps
Firmware Upgrade	◆ Main Serial Port Upgrade ◆ DFOTA Upgrade
Operating temperature	◆ Normal operating temperature range (3): -35~+75°C ◆ Extended operating temperature range (4): -40~+85°C
Storage temperature	Storage temperature range (5): -40~90°C
RoHS	All components comply with EU RoHS standards.

Remarks

(1) For a detailed description of the extended voltage, see section 6.2.

(2) Different firmware versions may support different network protocols; please refer to the documentation of the specific firmware version being used.

(3)(4)(5) Specific descriptions regarding temperature can be found in section 6.6.

2.3 Functional block diagram

The following diagram is the product functional block diagram:

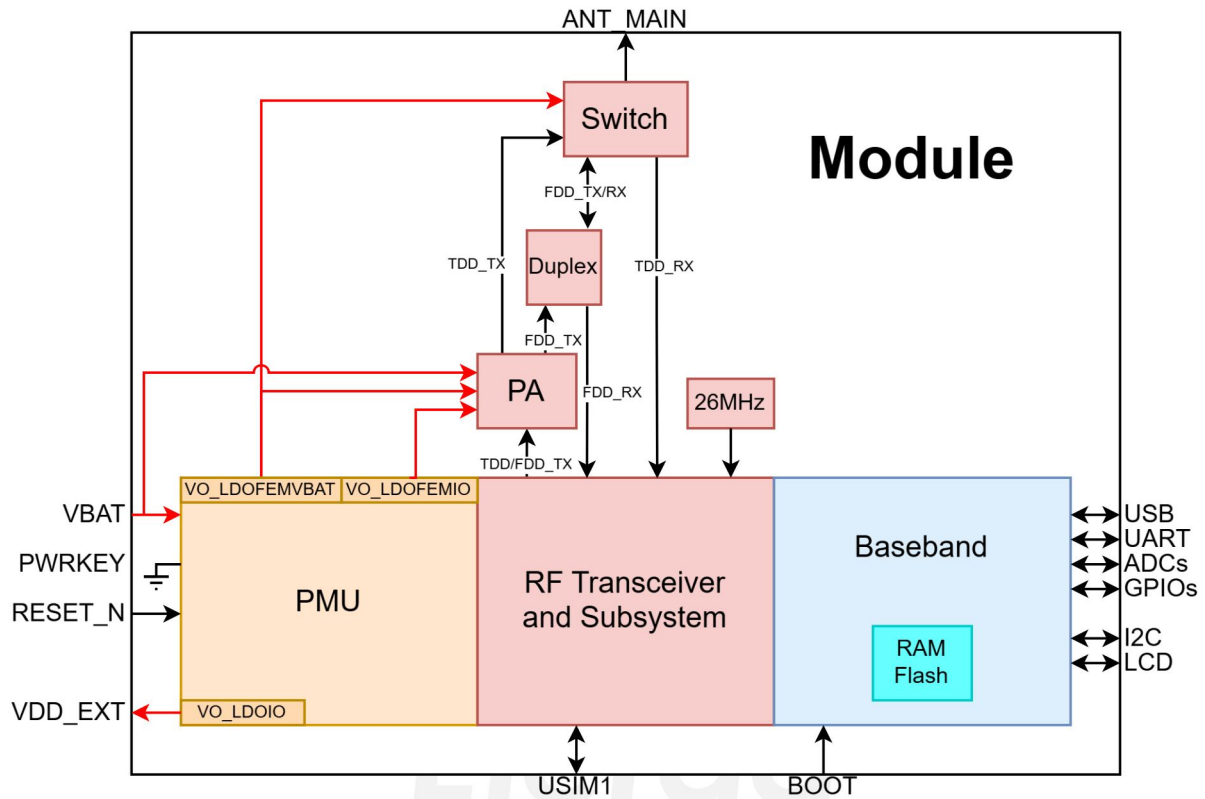


Figure 2-1 Functional Block Diagram

2.4 Pin distribution diagram

The NT26-FCN C series module has a total of 44 LCC pins:

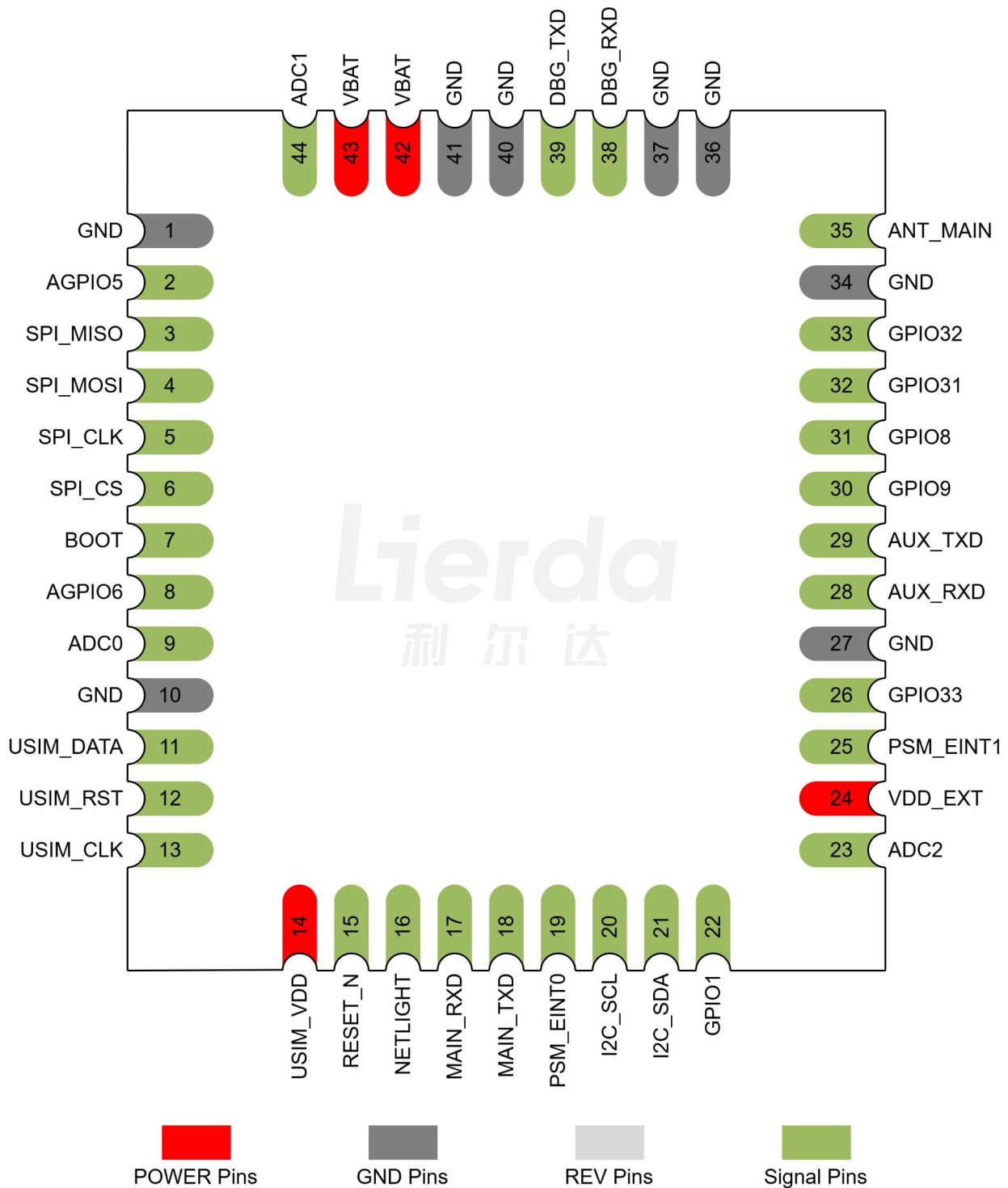


Figure 2-2 Module Pin Assignment Diagram

2.5 Pin Description Table

For better understanding of the application, the following table provides the type definitions of the I/O parameters:

Table 2-3 Type Definition Description of I/O Parameters

I/O Parameter Types	Explanation
DI	Digital input
DO	Digital output
DIO	Digital input and output
AI	Simulated input
AO	Simulated output
AIO	Simulated input and output
PI	Power input
PO	Power output
G	Grounding
OD	Drain open circuit

For better understanding and application, the following table provides a description of the characteristics of the power domain parameters:

Table 2-4 Module Power Domain Characteristics Description

Power domain parameters Type	DC Characteristics	Explanation	Power supply target
VBAT	Vmax=4.5V Vmin=2.3V Vnorm=3.6V	Module power input, it is recommended to use a 3.6V/1.2A power supply;	Module
VDD_EXT	VILmax=0.2×VDD_EXT VIHmin=0.7×VDD_EXT VOLmax=0.15×VDD_EXT VOHmin=0.8×VDD_EXT	Default 3.3V, configurable via software; Maximum drive capability 120mA; In sleep mode, VDD_EXT powers down, and the associated IO ports also	Reference level output

		power down simultaneously.	
VO_LDOIO	$V_{ILmax}=0.2 \times VO_LDOIO$ $V_{IHmin}=0.7 \times VO_LDOIO$ $V_{OLmax}=0.15 \times VO_LDOIO$ $V_{OHmin}=0.8 \times VO_LDOIO$	Default 3.3V, configurable via software; Maximum drive capability 120mA; VO_LDOIO powers down in sleep mode, and the associated IO ports also power down simultaneously.	VDD_EXT UART GPIO
VDD18AON	$V_{ILmax}=0.2 \times VDD18AON$ $V_{IHmin}=0.75 \times VDD18AON$	No power loss in sleep mode	RESET_N PSM_EINT0 PSM_EINT1
LDO_AONIO	$V_{ILmax}=0.2 \times LDO_AONIO$ $V_{IHmin}=0.7 \times LDO_AONIO$ $V_{OLmax}=0.15 \times LDO_AONIO$ $V_{OHmin}=0.8 \times LDO_AONIO$	Default 3.3V, software configurable; No power loss in sleep mode	AGPIO
VO_LDOSIM	$V_{norm}=1.8/3.0V$ $V_{OLmax}=0.15 \times VO_LDOSIM$ $V_{OHmin}=0.8 \times VO_LDOSIM$ $V_{ILmax}=0.2 \times VO_LDOSIM$ $V_{IHmin}=0.7 \times VO_LDOSIM$	Dedicated power supply for SIM cards, supporting 1.8/3.0V cards	USIM

Table 2-5 Module Pin Description

1. Power					
Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
42,43	VBAT	PI	Power supply	$V_{max}=4.5V$ $V_{min}=3.3V$ $V_{norm}=3.8V$	The external power supply needs to provide a current carrying capacity of more than 1.2A.
24	VDD_EXT	PO	Reference level output	$V_{norm}=3.3V$ $I_{omax}=120mA$	Can be used for external circuit pull-up or reference level; power off in sleep mode; leave floating if unused, it is recommended to reserve

					a test point.
1,10,27, 34,36,37 40,41	GND		GND		

2. Reset RESET_N

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
15	RESET_N	DI	Reset module	VDD18AON	Default internal pull-up, active low.

3. Wake up PSM_EINT

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
19	PSM_EINT0	DI	External Wake-up Module	VDD18AON	Default internal pull-up, active on falling edge
25	PSM_EINT1	DI	External wake-up module	VDD18AON	Default internal pull-up, active on falling edge

4. ADC Interface

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
9	ADC0	AI	General modular conversion	Voltage range: 0~1.5V	If not used, leave floating
23	ADC2	AI	General modular conversion		
44	ADC1	AI	General modular conversion		

5. Network Status Indicator*

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
16	NETLIGHT	DO	Network status indicator light	LDO_AONIO	If not used, leave floating

6. Serial Port UART					
Pin number	Pin name	I/O	Description	DC characteristics	Note
17	MAIN_RXD	DI	Main serial port: Module receiving data	VO_LDOIO	The main serial port supports low-power serial port wake-up. In sleep mode, the main serial port RXD does not lose power, while other serial ports will lose power. For details, see section 4.1.
18	MAIN_TXD	DO	Main serial port: Module sends data		
38	DBG_RXD	DI	Debugging serial port: Module receiving data		
39	DBG_TXD	DO	Debugging serial port: Module sends data		
28	AUX_RXD	DI	Application Serial Port: Module receiving data		
28	AUX_TXD	DO	Application serial port: Module sends data		

7. External USIM Card Interface					
Pin number	Pin name	I/O	Description	DC characteristics	Remarks
11	USIM_DATA	DIO	SIM card data cable	VO_LDOSIM	
12	USIM_RST	DO	SIM card reset line		
13	USIM_CLK	DO	SIM card clock line		
14	USIM_VDD	PO	SIM card power supply		

8. Antenna Interface RF

Pin number	Pin name	I/O	Description	DC characteristics	Remarks
35	ANT_MAIN	AIO	RF antenna interface	50Ω characteristic impedance	

9. General IO Interface

Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
7	BOOT	DIO	Internal connection GPIO0 Emergency Download Mode Control	VO_LDOIO	This pin is multiplexed as GPIO0, and pull-up is prohibited before power-on.
22	GPIO1	DIO	Internal connection GPIO1		If not used, leave floating
26	GPIO33	DIO	Internal connection GPIO33		If not used, leave floating
30	GPIO9	DIO	Internal connection GPIO9		If not used, leave floating
31	GPIO8	DIO	Internal connection GPIO8		If not used, leave floating
32	GPIO31	DIO	Internal connection GPIO31		If not used, leave floating
33	GPIO32	DIO	Internal connection GPIO32		If not used, leave floating
2	AGPIO5	DIO	Internal connection AGPIO5	LDO_AONIO	If not used, leave floating
8	AGPIO6	DIO	Internal connection AGPIO6		If not used, leave floating

10. SPI Interface*					
Pin number	Pin name	I/O	Description	DC Characteristics	Remarks
3	SPI_MISO	DI	SPI Master In Slave Out	VO_LDOIO	If not used, leave floating
4	SPI_MOSI	DO	SPI Master Out Slave In		If not used, leave floating
5	SPI_CLK	DO	SPI Serial Clock		If not used, leave floating
6	SPI_CS	DO	SPI Chip Select		If not used, leave floating

11. I2C Interface*					
Pin number	Pin Name	I/O	Description	DC Characteristics	Remarks
20	I2C_SCL	DO	I2C serial clock	VO_LDOIO	If not used, leave floating
21	I2C_SDA	DIO	I2C serial data		If not used, leave floating

2.6 Evaluation Kit

Lierda can provide complete evaluation and development kits, including development boards with USB interfaces. Please feel free to contact us for inquiries.

3 Work Characteristics

3.1 Work Mode

The NT26-FCN C series modules have multiple operating modes. The table below briefly describes several common operating modes of the modules:

Table 3-1 Working Modes

Mode	Status Description
Normal working mode	Data transmission: Network connection is normal. Module power consumption depends on network settings and data transmission rate.
	Idle state: The software is running normally. The module is connected to the network but has no data interaction with the network.
Minimum Function Mode	AT+CFUN=0 can set the module to minimum functionality mode, where both the RF and USIM card do not operate.
Sleep mode	The module's power consumption will be reduced to an extremely low level, but the module can still receive paging, short messages, and TCP/UDP data.

Remarks

1. For AT+CFUN related commands, please refer to the AT command data manual.
2. For detailed working modes and corresponding power consumption data, please refer to the PMU application documentation.

3.2 Sleep mode

The module can enter sleep mode through AT commands. In sleep mode, the module can reduce power consumption to a very low level, while the module's RF can still receive network information.

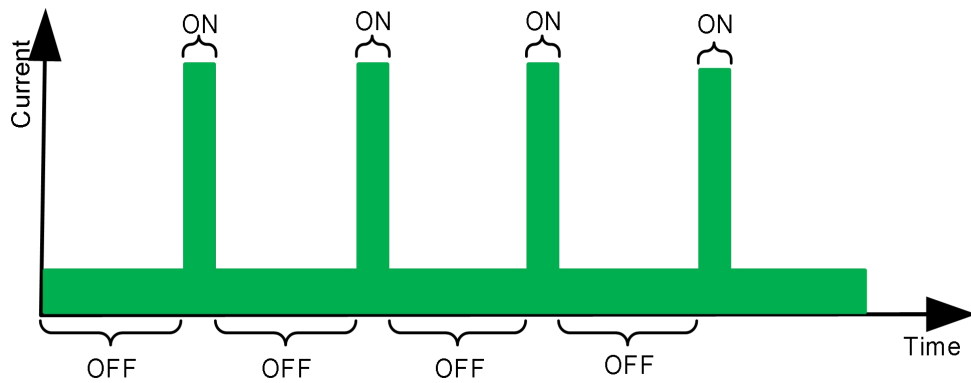


Figure 3-1 Power consumption states of the module in sleep mode

In sleep mode, the module peripherals are powered down to reduce power consumption, but the RF remains in intermittent reception mode and can still receive network information.

3.2.1 Main Serial Port Application Scenarios

If the module and the host communicate via UART, the following conditions must be met simultaneously for the module to enter sleep mode:

◆ Executing the AT+QSCCLK* command will enter sleep mode. For details, see the "AT Command Manual";

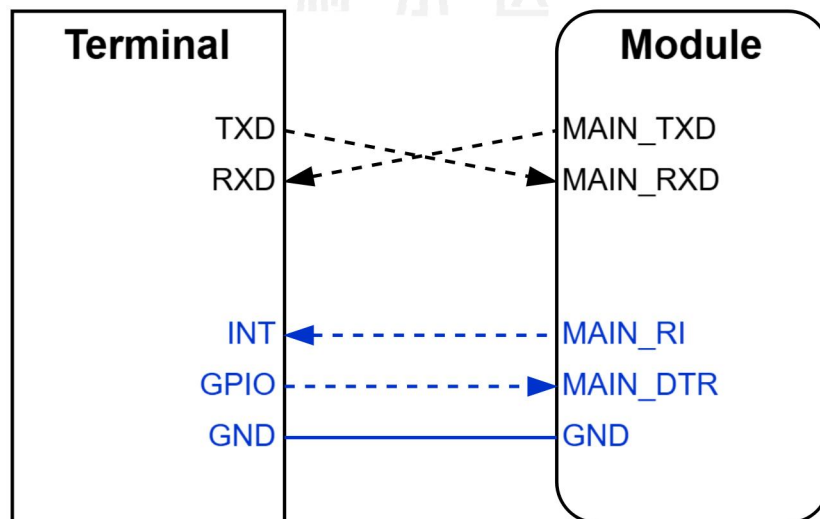


Figure 3-2 Application Circuit of Sleep Mode UART

- ◆ When the module has a URC to report, the module notifies the host through MAIN_RI*.
- ◆ The host can wake up the module by pulling MAIN_DTR low.

- ◆ If the host continuously pulls MAIN_DTR low, the module can be prevented from entering sleep mode.
- ◆ After entering sleep mode, the module can be awakened by sending any command to the main UART interface.

3.3 Power Supply Design

The NT26-FCN C series module provides two VBAT pins for connecting external power supplies. Additionally, VDD_EXT is the module's external output power supply and can be used as a reference voltage. The interface description is as follows:

Table 3-2 Power Pin Definitions

Pin number	Pin name	Description	Minimum value	Typical value	Maximum value	Unit
42,43	VBAT	Modular power supply	2.3	3.6	4.5	V
24	VDD_EXT	Provide reference level	-	3.3	-	V

The module-side VBAT power supply range is 2.3~4.5V. It is necessary to ensure that the voltage drop caused by the module's operation does not cause the minimum input voltage to fall below 2.3V. The following diagram shows the VBAT voltage drop situation of the module during burst transmission:

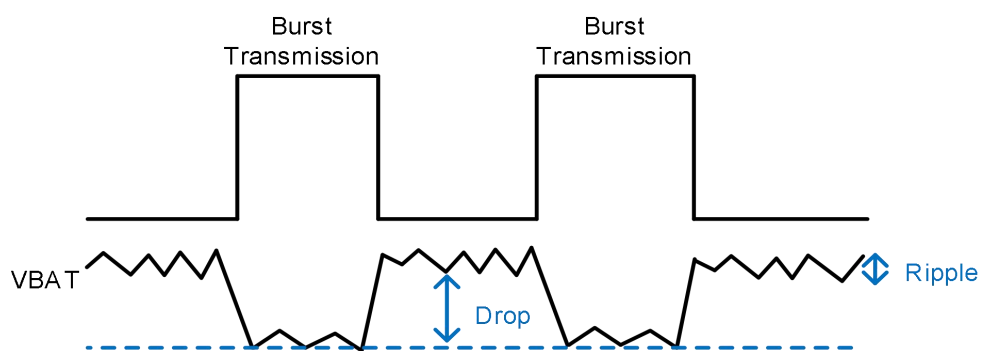


Figure 3-2 Burst Transmission Power Requirements

Remarks

In sleep mode, VDD_EXT will be powered off.

3.3.1 Power Supply Circuit Reference Design

When the module operates at the maximum transmission power of the 4G network, the transient operating current in the live network can reach 1.2A, which may cause a drop in the power supply voltage. Under any circumstances, it is necessary to ensure that the module's power supply voltage remains above 3.3V; otherwise, the module may experience unexpected issues such as rebooting. To reduce voltage drop, it is recommended to use a 100uF filter capacitor with low ESR ($ESR < 0.7\Omega$). It is also advised to reserve three chip-type multilayer ceramic capacitors (MLCC) with good ESR performance for VBAT, and the capacitors should be placed close to the VBAT pin. Additionally, to improve the module's surge voltage tolerance, a high-power TVS diode with low clamping voltage and high reverse pulse current (I_{pp}) should be added at the VBAT terminal.

The power supply reference circuit is as follows:

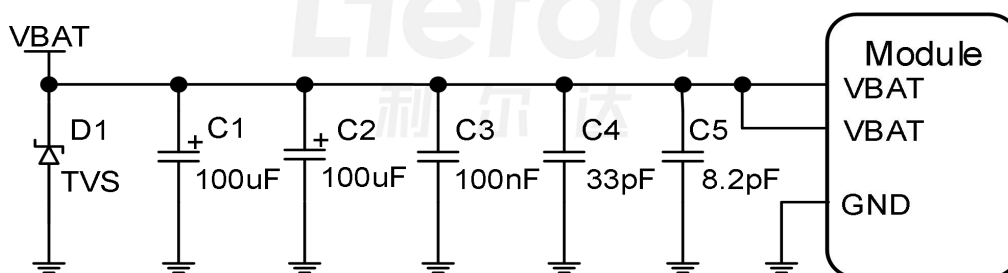


Figure 3-4 Module Power Supply Circuit Reference

In PCB design, the longer the VBAT trace, the wider the trace width should be. It is recommended that the trace width be no less than 2mm. The GND plane in the power section should be as complete as possible with multiple ground vias. At the same time, decoupling capacitors should be placed as close as possible to the VBAT pin of the module. Among them:

- ◆ D1 is a high-power TVS diode with a low clamping voltage;
- ◆ C1 and C2 are 100uF low ESR tantalum capacitors, enhancing the power supply's current sustainment capability and stabilizing the voltage;

◆ C3~C5 are filter capacitors with ceramic packages of 100nF, 33pF, and 8.2pF respectively, used to eliminate high-frequency interference.

3.4 Power on

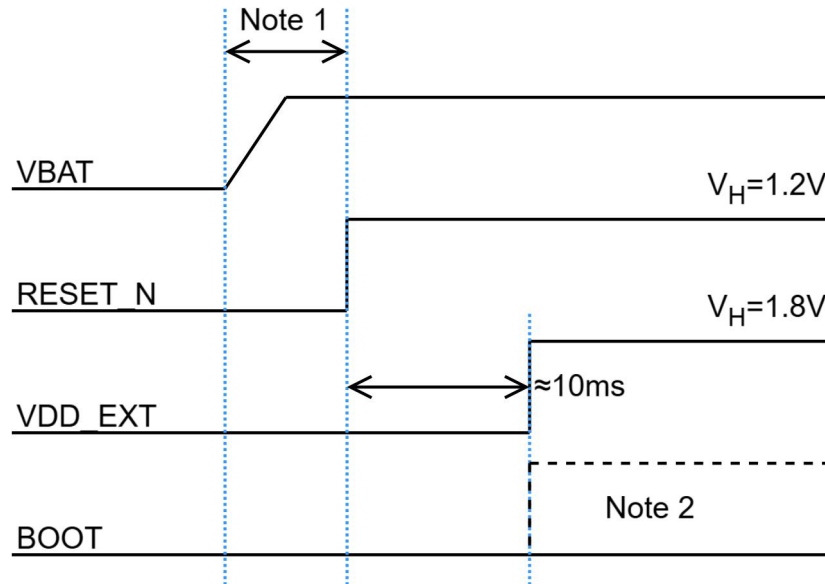


Figure 3-5 Power-Up Timing Sequence

Precautions

◆ The initial state of VBAT power-on must be less than 0.1V, and the time for VBAT to rise from 0V to 2.5V should be less than 10ms; before pulling the PWRKEY pin low, ensure that the VBAT voltage is stable. It is recommended that the time interval between powering on VBAT and pulling the PWRKEY pin low be no less than 30ms.

◆ The BOOT pin should be pulled to VDD_EXT before the module powers on; when the module powers on, it will enter emergency download mode.

◆ The main module serial port sends the character "Lierda" to indicate that the power-on is complete.

◆ When the module is operating normally, do not immediately cut off the module power, as this may damage the internal Flash data of the module. It is recommended to first enter flight mode via AT commands before disconnecting the power.

3.5 Reset

3.5.1 Reset Introduction

The module resets in the following way:

- ◆ Hardware reset: RESET_N pin, active low (pull low to reset, recommended duration greater than 50ms).
- ◆ Software reset: Send the "AT+ECRST" command to reset.

Table 3-3 Reset Pin Description

Pin number	Pin Name	I/O	Description	Remarks
15	RESET_N	DI	Reset module	Active low

When the external control of the module's RESET_N pin outputs a high level greater than 1.45V and less than 3.6V, and a low level less than 0.3V, the module's RESET_N pin can be controlled by direct connection. In other cases, an open-collector driver circuit must be used for control, as referenced below.

3.5.2 Circuit Reference Design

The RESET_N signal is relatively sensitive to interference. It is recommended that the module interface wiring be kept as short as possible and should be grounded. The typical reference hardware circuit applications are as follows. Two circuit diagrams can be selected according to actual needs:

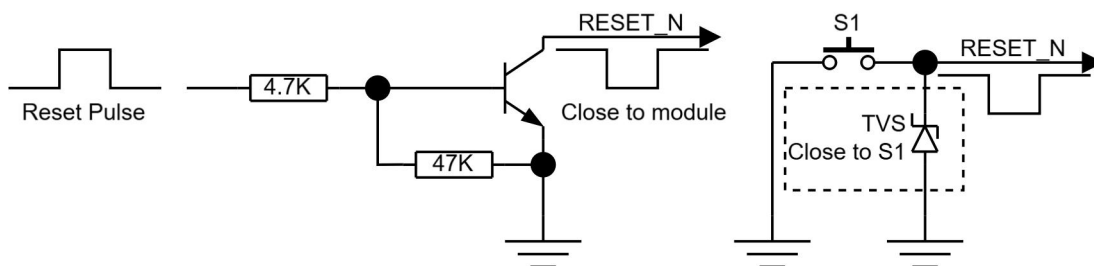


Figure 3-6 Reset Circuit Reference Design

3.5.3 Reset timing

The reset timing sequence of the module is as follows:

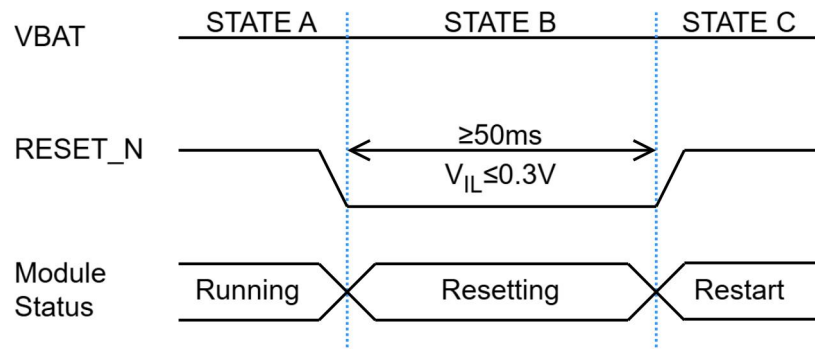


Figure 3-12 Reset Timing

Reset timing description:

- ◆ STATE A: Module normal working state;
- ◆ STATE B: Pull the RESET_N line low for at least 300ms;
- ◆ STATE C: The module begins to restart.

3.6 Wake up

The module, while in sleep mode, needs to be awakened through the PSM_EINT0 and PSM_EINT1 pins.

Table 3-4 Reset Pin Description

Pin number	Pin name	I/O	Description	Remarks
19	PSM_EINT0	DI	External Wake-up Module	Falling edge active
25	PSM_EINT1	DI	External Wake-up Module	Falling edge triggered

Precautions

- ◆ In addition to waking up via the PSM_EINT0 and PSM_EINT1 pins, the module can also be awakened through the low-power serial port. For details, see section 4.1.1.

3.6.1 Wake-up Circuit Reference Design

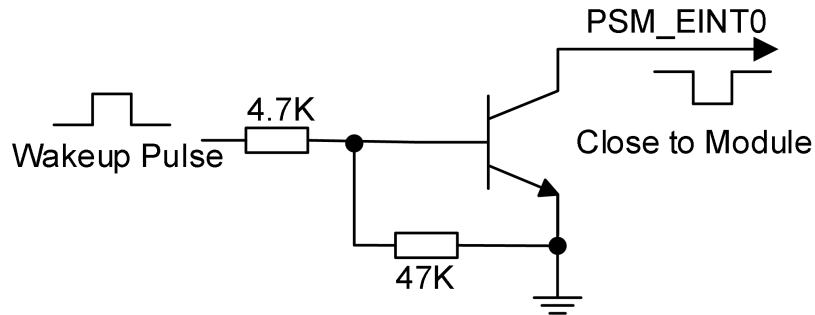


Figure 3-7 Transistor Wake-up Reference Circuit

Precautions

◆ The wake-up line wiring should not be too long, ensure ground shielding protection, keep it away from RF, VBAT power sources, and strong signal interference sources. The transistor should be placed as close as possible to the module's PSM_EINT0 and PSM_EINT1 pins to avoid external signal interference.

4 Application Interface

The NT26-FCN C series module provides the following application interfaces:

- ◆ 3-way UART interface
- ◆ 1-channel USIM interface
- ◆ 9-channel GPIO interface
- ◆ 1-channel SPI interface*
- ◆ 1-channel I2C interface*
- ◆ 3-channel ADC interface

4.1 UART Communication

The module provides three general asynchronous transceivers: the main serial port, the debug serial port, and the application serial port. The baud rate of the main serial port can be configured to 4800bps/9600bps/115200bps/921600bps (for firmware burning); the baud rate of the debug serial port is 3Mbps, used only for debugging; the application serial port can be used for communication with other peripherals.

Table 4-1 Serial Port Pin Definitions

Pin number	Pin name	I/O	Description	Remarks
17	MAIN_RXD	DI	Main serial port: Module receives data	MAIN_RXD switches the power domain in sleep mode: Non-sleep state: VO_LDOIO power domain; Sleep state: Switch to internal AON power domain
18	MAIN_TXD	DO	Main serial port: Module sends data	
38	DBG_RXD	DI	Debug serial port: Module receives data	
39	DBG_TXD	DO	Debug serial port: Module sends data	
28	AUX_RXD*	DI	Application serial port: Module receiving data	
29	AUX_TXD*	DO	Application serial port: Module sends data	

Precautions

- ◆ The serial port interface belongs to the VO_LDOIO power domain, with a default voltage level of 3.3V;
- ◆ The VO_LDOIO power domain will power down in sleep mode, and MAIN_RXD switches to the internal 1.2V power supply;
- ◆ The serial port RXD pin has an internal pull-up resistor, so no external pull-up resistor is needed, and it has reverse current protection.
- ◆ Pay attention to the consistency of voltage levels when using the serial port, otherwise leakage current may easily occur.

4.1.1 Serial port application

Main Serial Port Features:

- ◆ Used for AT command communication and data transmission, with a default baud rate of 9600bps.
- ◆ Configurable to 2400bps/4800bps/9600bps/115200bps.
- ◆ When using a baud rate higher than 9600 in sleep mode, the data received in the first packet will experience packet loss.
- ◆ Used for firmware upgrade, with a baud rate of 921600bps during the upgrade.
- ◆ The main serial port supports low-power serial port wake-up, and in sleep mode, RXD switches to the internal 1.2V power domain.
- ◆ The main serial port supports low-power serial port wake-up. Pulling RXD low in sleep mode will wake up the module, while VDD_EXT will be powered off in sleep mode, so VDD_EXT cannot be used for pull-up.

Debugging serial port characteristics:

- ◆ Using the dedicated tools provided by the platform, you can debug the serial port to view log information for software debugging, with a baud rate of 3 Mbps.

The schematic diagram of the main serial port connection method is as follows:

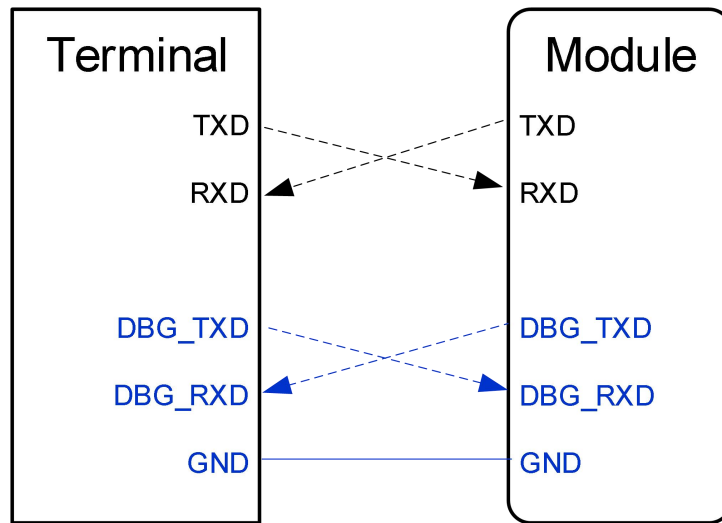


Figure 4-1 Serial Port Connection Diagram

4.1.2 Serial Port Circuit Reference Design

The main factors to consider for a suitable serial port level conversion circuit include whether it meets the operating speed of the serial port, whether it is suitable for low power consumption scenarios, and whether its power consumption meets the requirements. It is recommended to refer to the following level matching circuit:

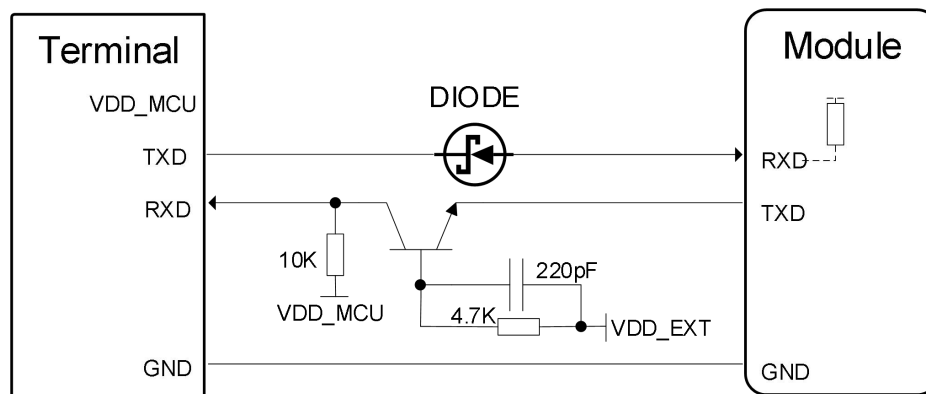


Figure 4-2 Level Conversion Reference Circuit

Recommended transistors for reference:

Brand: CJ Specification Model: S8050 J3Y Package: SOT-23

Recommended diode, please note that the forward voltage of the diode should be $\leq 0.3V$, for reference:

Brand: LRC Specification Model: LRB520S-30T1G Package: SOD-523

4.2 USIM card interface

The module includes one external USIM card interface, supporting module access to the USIM card. This USIM card interface complies with 3GPP specifications. The external USIM card is powered by the internal power supply of the module and supports cards with 1.8V/3.0V power supply.

Table 4-2 External USIM Card Interface Pin Definitions

Pin number	Pin name	I/O	Description	Remarks
11	USIM_DATA	IO	SIM card data cable	
12	USIM_RST	DO	SIM card reset line	
13	USIM_CLK	DO	SIM card clock line	
14	USIM_VDD	PO	SIM card power supply	Supports 1.8V/3V USIM cards when $3.0V \leq VBAT \leq 4.5V$. When $2.3V \leq VBAT \leq 3V$, only 1.8V USIM cards are supported.

4.2.1 USIM Card Circuit Reference Design

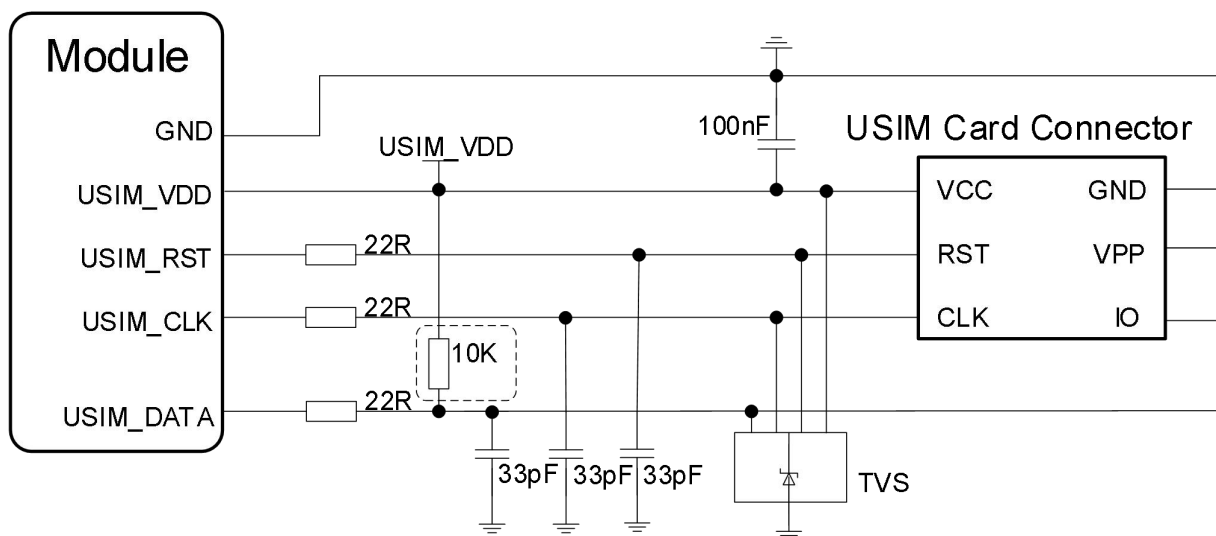


Figure 4-3 6PIN External SIM Card Reference Circuit

4.2.2 Precautions for USIM Card Circuit Design

To ensure the reliability and availability of the SIM card in the application, please follow the following USIM circuit design guidelines:

- ◆ When laying out, try to place the SIM card as close to the module as possible, and keep the signal line routing length within 200mm if possible;
- ◆ The SIM card signal line should be kept away from the RF routing and VBAT power line.
- ◆ The GND wiring of the SIM card should be short and thick, ensuring the wiring width is not less than 0.5mm;
- ◆ The decoupling capacitor for USIM_VDD should not exceed 1uF, and the capacitor should be placed close to the external SIM card's VCC.
- ◆ To avoid signal crosstalk between USIM_DATA and USIM_CLK, the routing of these two lines should not be too close. A ground shield must be added between the two traces. Additionally, to prevent the effects of excessively long traces, USIM_DATA generally requires a pull-up resistor to USIM_VDD to enhance driving capability. If the trace is too long, it is recommended to externally reserve a 10K resistor placed near the card slot. Furthermore, the USIM_RST signal also requires ground shielding protection.
- ◆ To ensure good ESD protection performance, it is recommended to add a TVS diode to the pins of the external USIM card. The parasitic capacitance of the selected TVS diode should not exceed 15pF; a 22-ohm resistor should be connected in series on the signal line between the module and the SIM card to suppress stray EMI and enhance ESD protection. At the same time, a parallel 33pF capacitor is used to filter out RF interference. The related resistors, capacitors, and TVS diodes should be placed close to the USIM card socket.

4.3 General IO Interface

The module provides 9 GPIO interfaces, with the interface description as follows:

Table 4-3 GPIO Pin Definitions

Pin number	Pin name	I/O	Description	Remarks
7	BOOT	DIO	Internal connection GPIO0 Emergency Download Mode Control	Pull-up is prohibited before power-on; if not used, leave it floating.
22	GPIO1	DIO	Internal connection GPIO1	If not used, leave floating
26	GPIO33	DIO	Internal connection GPIO33	If not used, leave floating
30	GPIO9	DIO	Internal connection GPIO9	If not used, leave floating
31	GPIO8	DIO	Internal connection GPIO8	If not used, leave floating
32	GPIO31	DIO	Internal connection GPIO31	If not used, leave floating
33	GPIO32	DIO	Internal connection GPIO32	If not used, leave floating
2	AGPIO5	DIO	Internal connection AGPIO5	If not used, leave floating
8	AGPIO6	DIO	Internal connection AGPIO6	If not used, leave floating

4.4 SPI Interface*

The module provides one SPI interface, with the interface description as follows:

Table 4-4 SPI Pin Definitions

Pin number	Pin name	I/O	Description	Remarks
3	SPI_MISO	DI	SPI Master In Slave Out	If not used, leave floating
4	SPI_MOSI	DO	SPI Master Out Slave In	If not used, leave floating
5	SPI_CLK	DO	SPI Serial Clock	If not used, leave floating
6	SPI_CS	DO	SPI Chip Select	If not used, leave floating

4.5 I2C interface*

The module provides one I2C interface, with the interface description as follows:

Table 4-5 I2C Pin Definitions

Pin number	Pin name	I/O	Description	Remarks
20	I2C_SCL	DO	I2C Serial Clock	If not used, leave floating
21	I2C_SDA	DIO	I2C serial data	If not used, leave floating

4.6 ADC interface

The module provides 3 ADC interfaces, with the interface descriptions as follows:

Table 4-6 ADC Interface Pin Definitions

Pin number	Pin name	I/O	Description	Remarks
9	ADC0	AI	General modular conversion	Internal direct connection
23	ADC2	AI	General modular conversion	Input voltage range: 0~1.5V
44	ADC1	AI	General modular conversion	Internal voltage division Input voltage range: 0~3.3V Default 1.5V, internal voltage division can be enabled via software.

4.7 Network status indicator light*

The NETLIGHT signal can be used to indicate the network status of the module.

The reference connection circuit for the network status indicator light is shown in the figure below:

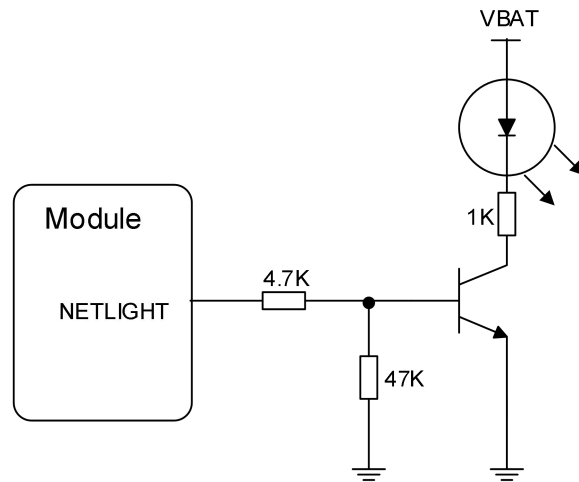


Figure 4-4 Network Indicator Light Interface Reference Circuit

5 Radio Frequency Characteristics

5.1 Introduction to Radio Frequency Functions

The NT26-FCN C series module provides a main antenna RF interface, through which all air interface data is transmitted or received. The interface characteristic impedance is 50 Ω . The interface description is as follows.

Table 5-1 RF Interface Pin Definitions

Pin number	Pin Name	I/O	Description	DC Characteristics	Remarks
35	ANT_MAIN	AIO	Main antenna interface	-	Used for over-the-air data transmission

5.2 Cellular network

5.2.1 Operating frequency band

The NT26-FCN C series modules support the following operating frequency bands:

Table 5-2 Operating Frequency Bands

Operating frequency band	Transmission (MHz)	Reception (MHz)
LTE-FDD B1	1920~1980	2110~2170
LTE-FDD B3	1710~1785	1805~1880
LTE-FDD B5	824~849	869~894
LTE-FDD B8	880~915	925~960
LTE-TDD B34	2010~2025	2010~2025
LTE-TDD B38	2570~2620	2570~2620
LTE-TDD B39	1880~1920	1880~1920
LTE-TDD B40	2300~2400	2300~2400
LTE-TDD B41	2535~2675	2535~2675

Precautions

LTE-TDD B41 only supports 140M (2535-2675MHz).

5.2.2 Transmission power

The transmission power of the NT26-FCN C series modules is as follows:

Table 5-3 RF Conducted Emission Power

Frequency band	Maximum value	Minimum value	Remarks
LTE-FDD B1 (10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	Compliant with Cat.1 protocols in 3GPP Rel-13 and Rel-14.
LTE-FDD B3 (10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	
LTE-FDD B5 (10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	
LTE-FDD B8 (10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	
LTE-TDD B34 (10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	
LTE-TDD B38 (10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	
LTE-TDD B39(10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	
LTE-TDD B40 (10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	
LTE-TDD B41 (10 MHz)	23dBm $\pm$ 2.7dB	<-40dBm	

5.2.3 Receiving sensitivity

The receiving sensitivity of the NT26-FCN C series module is as follows:

Table 5-4 Conduction Reception Sensitivity under Single Transmission (Throughput $\geq$ 95%)

Frequency band	Reception Sensitivity (10M)	3GPP Standard (10M)
LTE-FDD B1 (10 MHz)	-98.3dBm	-96.3dBm
LTE-FDD B3 (10 MHz)	-98.8dBm	-93.3dBm
LTE-FDD B5 (10 MHz)	-98.8dBm	-94.3dBm

LTE-FDD B8 (10 MHz)	-98.9dBm	-93.3dBm
LTE-TDD B34 (10 MHz)	-101.8dBm	-96.3dBm
LTE-TDD B38 (10 MHz)	-99.7dBm	-96.3dBm
LTE-TDD B39(10 MHz)	-101.7dBm	-96.3dBm
LTE-TDD B40 (10 MHz)	-100.5dBm	-96.3dBm
LTE-TDD B41 (10 MHz)	-99.8dBm	-94.3dBm

5.3 Antenna Reference Circuit Design

When users are using the NT26-FCN C series module, a π -type matching circuit needs to be added between the module's RF antenna interface and the user's baseboard antenna interface. The typical antenna matching circuit and initial parameters are shown in the figure below:

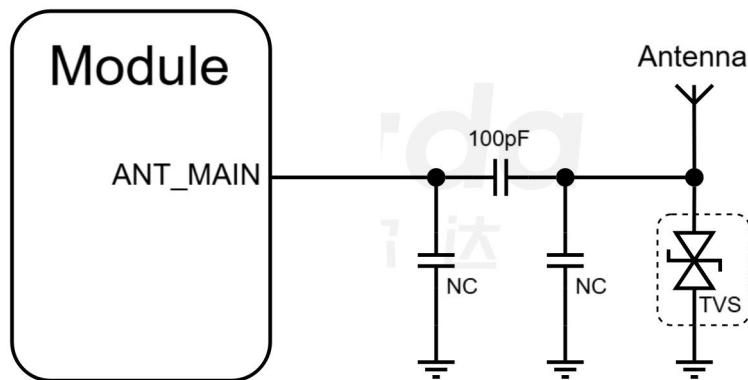


Figure 5-1 RF Antenna Reference Circuit

Precautions

◆ The series position uses 100pF, the capacitor position is default not mounted, and the component package is recommended to use 0201 or 0402 package.

◆ If using an external antenna, or if the user can touch the antenna, it is recommended to reserve a TVS diode to enhance electrostatic protection. Since the parasitic capacitance of the TVS may affect antenna performance, a model with parasitic capacitance $\leq 0.5\text{pF}$ should be selected. After adding the TVS diode, it is recommended to

retune the antenna.

5.4 Guidelines for RF Signal Line Routing

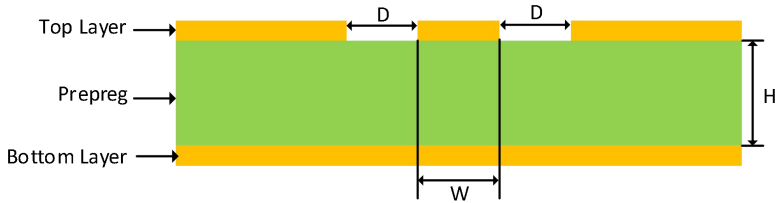
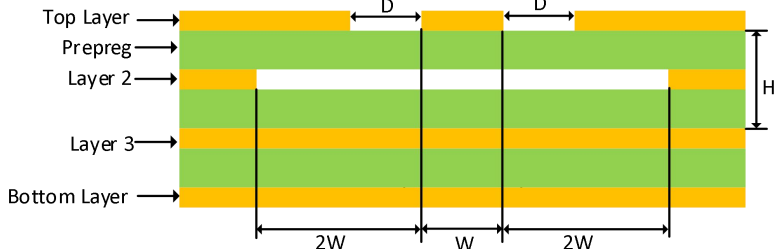
The routing between the module antenna interface and the user antenna must comply with the 50-ohm characteristic impedance requirement. At the same time, the RF routing should be as short as possible to ensure that the insertion loss of the RF routing is minimized.

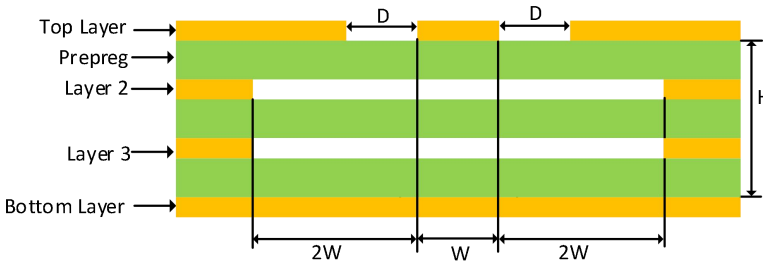
Under normal circumstances, the impedance of the RF signal line is determined by the dielectric constant ϵ_r of the material, the trace width W , the distance to ground D , and the thickness H of the reference ground plane.

In the field of Internet of Things applications, the design of PCB characteristic impedance is usually implemented using the coplanar waveguide method, which helps achieve better shielding of RF signal lines and enables higher integration for compact designs.

Common PCB coplanar waveguide designs include the following types:

Table 5-5 Common PCB Coplanar Waveguide Structure Design Description

Serial Number	Pin Name	Remarks
1	 Top Layer → Prepreg → Bottom Layer → Dimensions: D, D, W, H	2-layer board The reference plane is the L2 layer.
2	 Top Layer → Prepreg → Layer 2 → Layer 3 → Bottom Layer → Dimensions: D, D, W, H, $2W$, W, $2W$	4-layer board The reference location is the L3 layer. L2 layer hollowing

3		4-layer board The reference plane is the L4 layer. L2 and L3 layer following
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During the layout process, a 50-ohm impedance can be simulated and calculated using the Polar Si9000 tool. Based on the actual situation, select the appropriate line width, line spacing, and stacking to achieve the best design effect.

Taking the calculation of a 2-layer board with a 1.6mm thickness PCB coplanar waveguide as an example:

- ◆ Select the appropriate PCB substrate. Once the substrate is chosen, the dielectric constant ϵ_r is basically determined (4.6 in this example);
- ◆ Choose an appropriate trace width. A good method is to select a trace width that matches the size of the pads of the components used (note that it should not be too thin; in this example, 0.65mm is chosen).
- ◆ Based on the selected board thickness, line width, dielectric constant, etc., the spacing to ground can be roughly calculated. If the line spacing is not appropriate, the line width can be adjusted to modify the line spacing (calculated as 0.14mm, which is basically suitable).

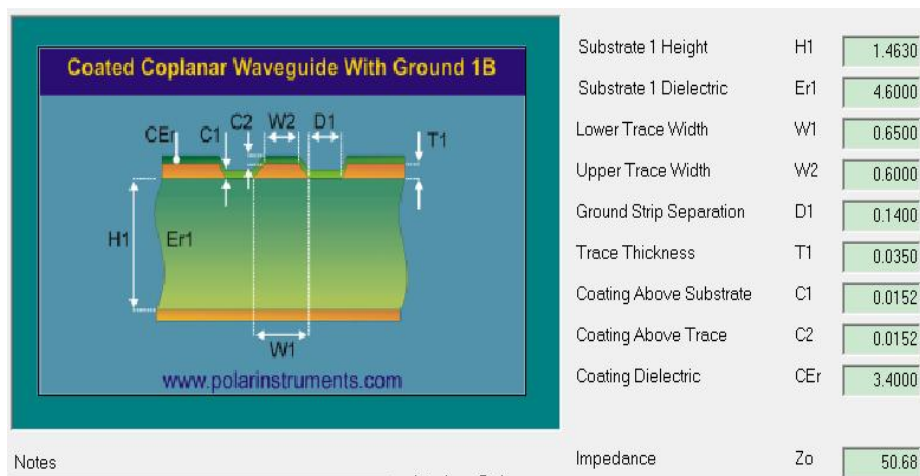


Figure 5-2 Reference for the calculation method of 50-ohm impedance

During the actual layout process, the following suggestions are provided for reference:

- ◆ In the π -type circuit, three matched reserved components are placed close together near the antenna, while bypass components are recommended to be placed on both sides of the RF line.
- ◆ Place irregular vias on both sides of the RF trace GND plane; there must be a complete GND plane underneath the entire RF trace area.
- ◆ There should be no other wiring beneath the RF cable to avoid affecting the RF performance or other circuits.

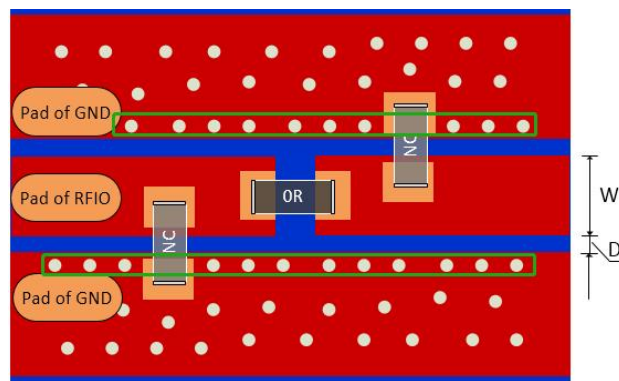


Figure 5-3 RF Trace Layout Design Diagram

The rationality of RF routing can be directly reflected in the module's conduction test. To ensure the product's overall performance is maximized, coordination with antenna design is also required. To better meet the needs of antenna design, the following requirements are hoped to be achieved in PCB design. Below are guidelines for PCBs with different numbers of layers:

- ◆ When the product PCB is designed with 2 layers, it is best for the TOP and BOTTOM LAYER directly under the module to both be GND layers. The routing lines that need to be led out from the module should avoid passing directly underneath the module and should be routed from the outside of the module.
- ◆ When the product PCB is designed with 4 layers, it is recommended that the module's signal traces be routed on the third or fourth layer, reserving the first and second layers as complete GND reference layers for the module.

5.5 Antenna Design Requirements

The radio frequency performance of the module is also affected by the antenna. The selection of the antenna needs to meet the following requirements:

- ◆ Select an antenna that matches the module's operating frequency band;
- ◆ The antenna's characteristic impedance is required to be 50 ohms to reduce the loss at the connection between the RF line and the antenna;
- ◆ The insertion loss within the working frequency band should be as low as possible, such as a voltage standing wave ratio (VSWR) ≤ 2 and return loss ≥ 10 dB.

Commonly used antennas suitable for Cat.1 scenarios are shown in the figure below:

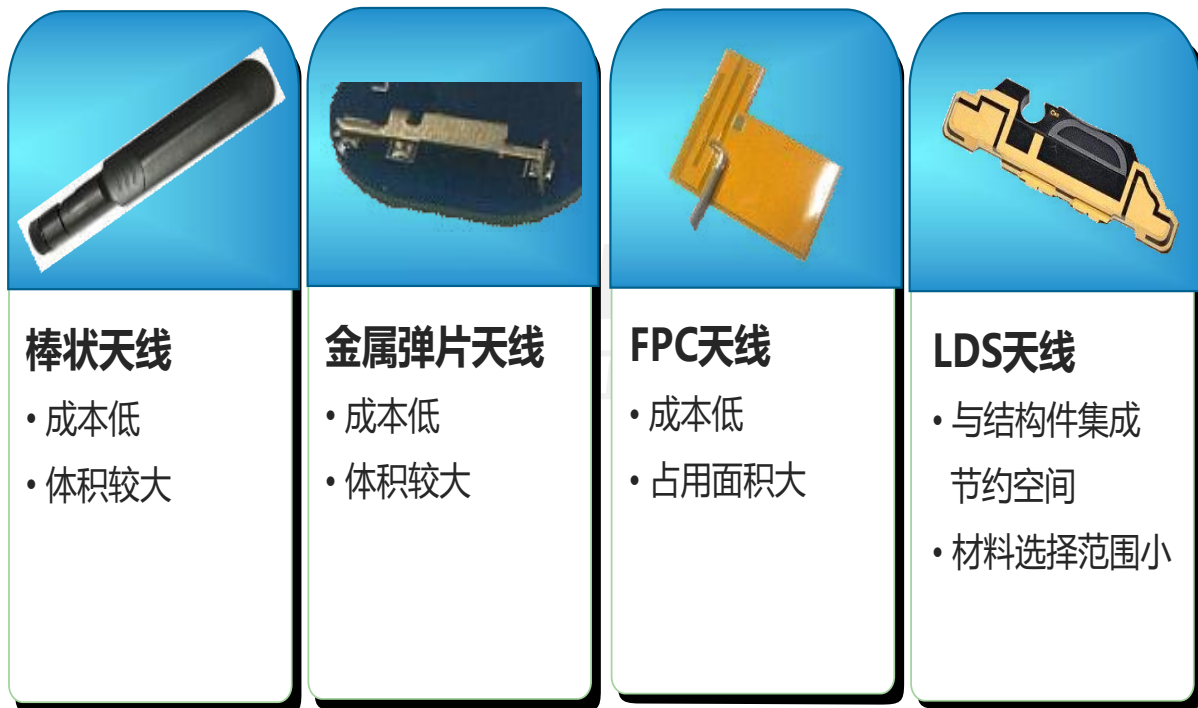


Figure 5-4 Cat.1 Common Antenna Types

5.6 Recommended RF Connectors

In some applications requiring customized antennas, such as when the device is shielded by an iron casing, an IPX or SMA connector can be reserved on the motherboard. The antenna can then be connected to the outside of the device via an RF cable to achieve better performance.

Table 5-6 Common RF Connectors

Serial Number	Product Name	Image	Remarks
1	IPX socket		Connect the RF cable
2	SMA connector		Can connect to RF cable or antenna

6 Electrical Performance and Reliability

6.1 Absolute maximum rating

Table 6-1 Maximum Rated Pin Voltages

Parameters	Description	Minimum value	Maximum value	Unit	Remarks
V _{BAT}	Module power supply voltage	-0.3	5.0	V	
V _{IAIO}	ADC input voltage	-0.3	3.6	V	
V _{IGPIO}	GPIO input voltage	-0.3	3.6	V	
Others	Other pin input voltage	-0.3	3.6	V	

Precautions

Exceeding the absolute maximum rated values during use may cause permanent damage to the module.

6.2 Power rating

Table 6-2 Operating Voltage Range

Parameters	Minimum value	Typical value	Maximum value	Unit	Remarks
Normal operating voltage (1)	3.1	3.6	4.5	V	
Extended operating voltage (2)	2.3	3.6	4.5	V	
USIM_VDD	-	1.8/3.0	-	V	
VDD_EXT ⁽³⁾	-	3.3	-	V	

Precautions

◆ When the module operates within this voltage range, the relevant performance of the module meets the 3GPP standard requirements.

◆ When the module operates within this voltage range, it can still maintain normal operation without irreversible failures; only certain parameters, such as output power, may exceed the 3GPP standard range. When the voltage returns to the normal operating range, all module parameters still comply with the 3GPP standards.

◆ Sleep Mode: VDD_EXT will power down in sleep mode.

6.3 Power consumption

The power consumption of the NT26-FCN C series module segment is as follows:

Table 6-4 Module Power Consumption

Module Description	Test conditions	Typical value	Unit
Shutdown mode	Module Shutdown*	-	μ A
Sleep mode	Hibernate	3.14	μ A
	Sleep2	13.86	μ A
	Sleep1	52.28	μ A
	LTE-FDD @ PF = 32	0.668	mA
	LTE-FDD @ PF = 64	0.36	mA
	LTE-FDD @ PF = 128	0.273	mA
	LTE-FDD @ PF = 256	0.142	mA
	LTE-TDD @ PF = 32	0.697	mA
	LTE-TDD @ PF = 64	0.378	mA
	LTE-TDD @ PF = 128	0.211	mA
	LTE-TDD @ PF = 256	0.142	mA
Idle mode	LTE-FDD @ PF = 64	3.47	mA
	LTE-TDD @ PF = 64	3.49	mA
LTE Data Transmission	LTE-FDD B1	583	mA
	LTE-FDD B3	587	mA
	LTE-FDD B5	579	mA
	LTE-FDD B8	525	mA

	LTE-TDD B34	189	mA
	LTE-TDD B38	215	mA
	LTE-TDD B39	201	mA
	LTE-TDD B40	272	mA
	LTE-TDD B41	228	mA

Precautions

◆ Module without power-off state;

◆ The typical power consumption values are based on the standard firmware.

Customized firmware may have different power consumption compared to the standard firmware due to enabled features. Actual power consumption should be measured on-site.

6.4 Digital Logic Electrical Characteristics

Table 6-5 Digital IO Logic Level Characteristics

Parameters	Description	Minimum value	Maximum value	Unit
VIH	Input high level	$0.7 \times VDD_EXT$	-	V
VIL	Input low level	-	$0.2 \times VDD_EXT$	V
VOH	Output high level	$0.8 \times VDD_EXT$	-	V
VOL	Output low level	-	$0.15 \times VDD_EXT$	V

Precautions

The module's GPIO reference level follows VDD_EXT. When VDD_EXT powers down during sleep, the corresponding GPIO will also power down.

The AGPIO logic level definitions for the NT26-FCN C series modules are as follows:

Table 6-6 AGPIO Logic Level Description

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	VIL	Input low level	-	$0.2 \times LDO_AONIO$	V
	VIH	Input high level	$0.7 \times LDO_AONIO$	-	V

Output	VOL	Output low level	-	$0.15 \times \text{LDO_AONIO}$	V
	VOH	Output high level	$0.8 \times \text{LDO_AONIO}$	-	V

Precautions

The module's AGPIO reference level follows LDO_AONIO and will not lose power during sleep mode.

The logic level definitions for the USIM card interface of the NT26-FCN C series module are as follows:

Table 6-7 USIM Logical Electrical Characteristics

Type	Parameters	Description	Minimum value	Maximum value	Unit
Input	VIL	Input low level	-	$0.2 \times \text{USIM_VDD}$	V
	VIH	Input high level	$0.7 \times \text{USIM_VDD}$	-	V
Output	VOL	Output low level	-	$0.15 \times \text{USIM_VDD}$	V
	VOH	Output high level	$0.8 \times \text{USIM_VDD}$	-	V

Precautions

The voltage level of USIM_VDD is determined based on the detected type of USIM card, supporting 1.8V/3.0V USIM cards.

6.5 Electrostatic Protection

In daily life and production, static electricity is everywhere. Static electricity generated by the human body and friction between objects can be transmitted to modules through various means and may potentially damage the modules. Therefore, it is essential to pay attention to static electricity protection and implement static protection measures. For example:

- ◆ Wear anti-static gloves during research and development, production, assembly, and testing processes;

◆ When designing products, add anti-static protection devices at circuit interfaces and other points susceptible to electrostatic discharge.

The electrostatic discharge performance of the module is as follows:

Table 6-8 Electrostatic Protection Characteristics

Test interface	Contact discharge	Remarks
VBAT	$\pm 8\text{KV}$	Test Standard: IEC61000-4-2 Temperature: 25°C Humidity: 45%
GND	$\pm 8\text{KV}$	
Antenna interface	$\pm 8\text{KV}$	
Others	-	

For detailed electrostatic protection design, refer to the "Lierda NT26-FCN C Series Hardware Reference Design Manual."

6.6 Operating and storage temperature

Table 6-9 Operating Temperature Range

Parameters	Minimum value	Typical value	Maximum value	Unit	Remarks
Normal operating temperature (1)	-35	+25	+75	°C	
Extended Operating Temperature (2)	-40	+25	+85	°C	
Storage ambient temperature (3)	-40	+25	+90	°C	

Precautions

(1)When the module operates within this temperature range, the module's relevant performance meets the 3GPP standard requirements.

(2)When the module operates within this temperature range, it can still maintain normal working status without any irrecoverable failures; only a few parameters, such as output power, may exceed the 3GPP standard range. When the temperature returns to the normal operating range, all module indicators still comply with the 3GPP standards.

(3)This storage temperature range does not include packaging materials; attention

should be paid to the maximum tolerable temperature of the tape packaging.

6.7 Precautions

There are some precautions to be aware of during the production and use of the module:

- ◆ When spraying the module, please try to avoid spraying material into the inside of the module;
- ◆ When cleaning the module, do not use ultrasonic cleaning on the module, as it may cause damage to the internal crystal of the module;
- ◆ When producing and using the module, please avoid applying it in environments or packaging containing any amount of mercury or mercury vapor, as this may lead to product failure or malfunction.

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7 Mechanical dimensions

7.1 Mechanical dimensions

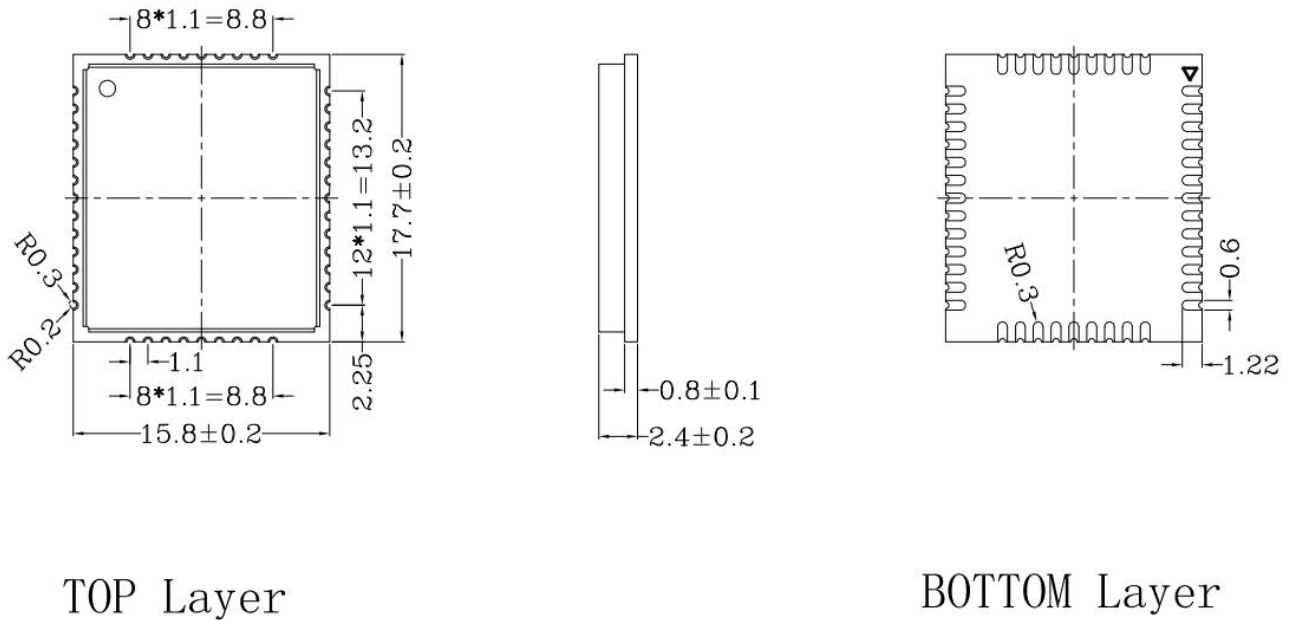


Figure 7-1 Module Mechanical Dimension Drawing

7.2 Module top view/bottom view



Figure 7-2 Module Top/Bottom View

The above are the design renderings of the module; please refer to the actual module for accuracy, especially as the label content is for illustration purposes only. For more information, such as module packaging recommendations, production guidelines, and

packaging methods, please refer to our company's production guide documents.

7.3 Recommended Packaging

The recommended module solder pads are shown in the figure below. Users can make fine adjustments according to their own production processes.

◆ The module's peripheral pins are designed with right angles internally. When designing the baseboard pads, users should consider using rounded corner transitions. For the square pads on the bottom of the module, the baseboard design can use the module pin dimensions, as shown in the reference design diagram of a single pad below.

◆ To facilitate the opening of stepped stencils, it is recommended not to place other components within 2.0mm outside the module pads. Users can refer to the requirements of their own stencil manufacturers to determine this distance.

Note: The module pads are symmetrically distributed around the module's center point.

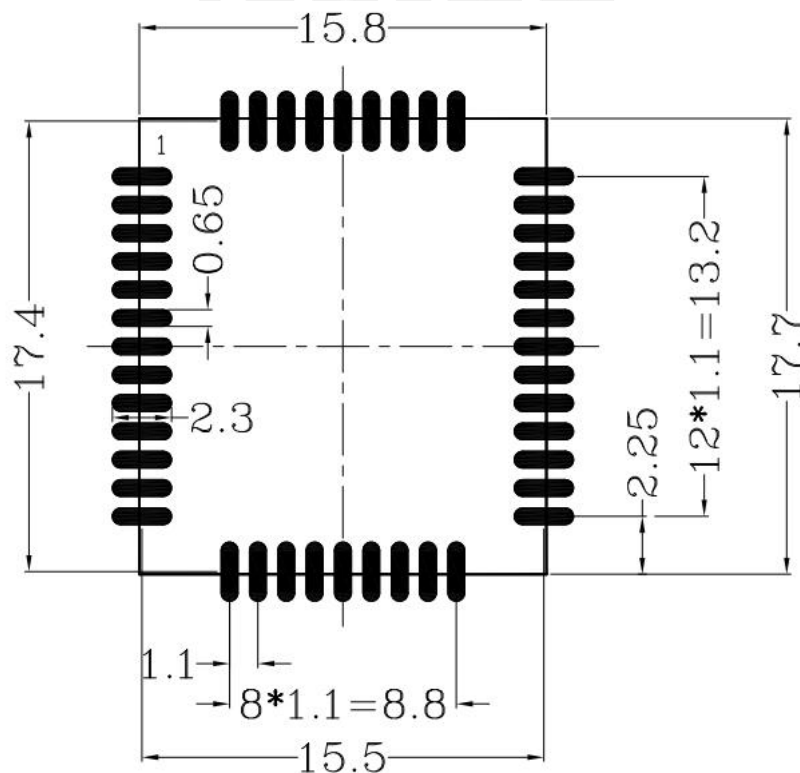


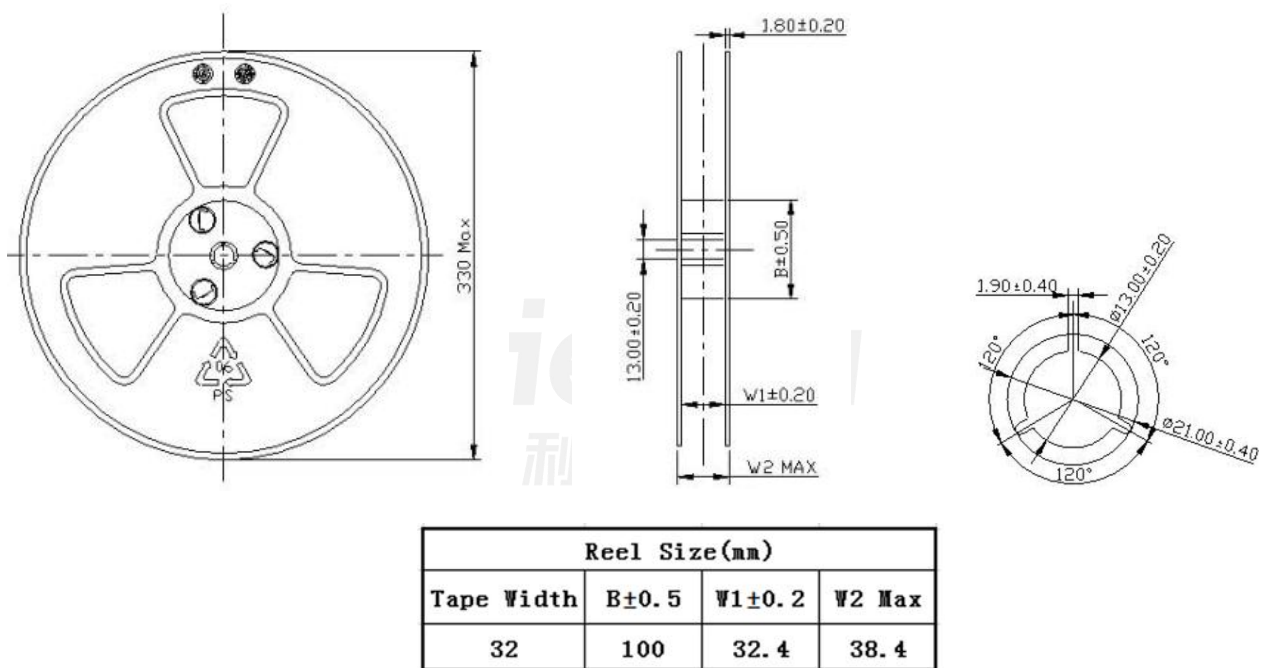
Figure 7-3 Recommended Module Pad

8 Packaging and Production Information

This chapter describes guidance information on the packaging, storage, production, and maintenance of the module, applicable to the assembly process guidance of the module.

8.1 Packaging specifications

The module's factory packaging uses a rubber wheel carrier tape method, with the rubber wheel reference dimensions as follows:



The direction of tape feeding is as follows: (Refer to the diagram, the label content is subject to the actual product, and pay attention to the position of module PIN1)

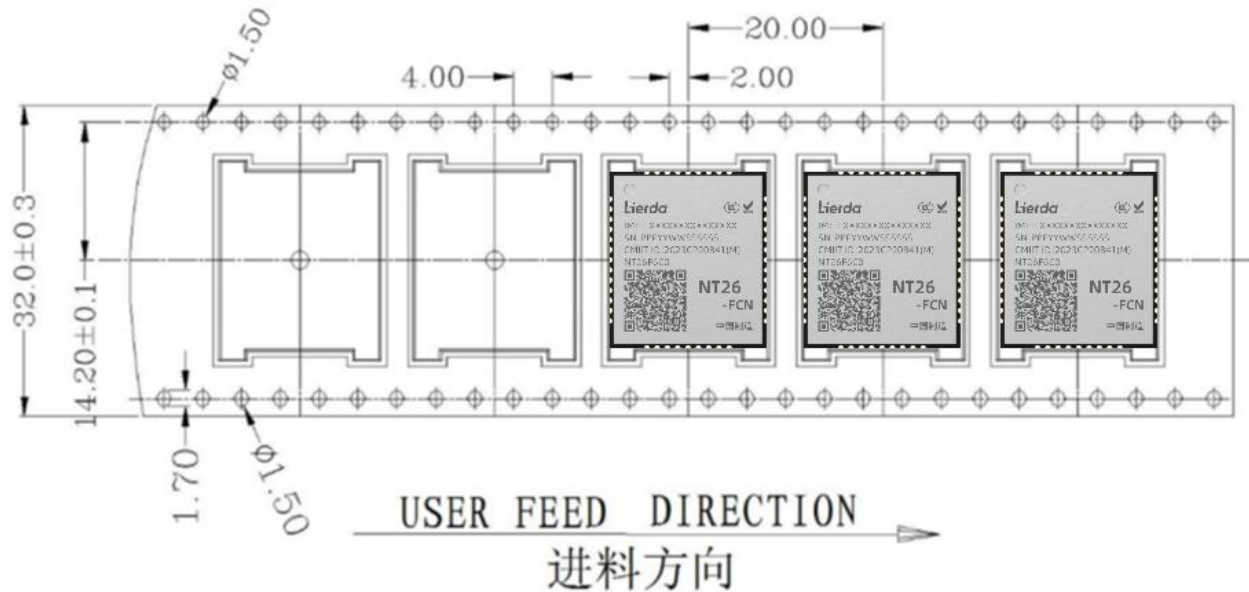


Figure 8-1 Packaging Specifications and Dimensions

8.2 Storage conditions

The module is shipped in the form of a vacuum reel-sealed bag, with a moisture sensitivity level of MSL 3.

Storage conditions:

- 1) When the temperature is below 40°C and the humidity is below 90% (RH), solderability can be ensured for 12 months under well-sealed packaging conditions.
- 2) After opening, ensure that the chip mounting is completed within 168 hours under the conditions of an ambient temperature below 30°C and a relative humidity below 60% (RH).

If the above conditions are not met, baking is required:

- 1) Tape packaging, baked at 60°C ± 5°C for 24~48 hours.
- 2) If accelerated baking is required, the module needs to be taken out from the tape reel and placed on a high-temperature resistant container (such as a tray) for baking (ESD protection must be observed during removal). Bake at 125°C ± 5°C for 8 hours.
- 3) The cumulative baking time must not exceed 96 hours.

For more detailed guidance, please refer to the IPC/JEDEC J-STD-033 standard.

8.3 Production welding

8.3.1 Reflow method

If the customer's module baseboard is a double-sided board, it is recommended to place the module during the second solder paste application. During the first solder paste application, it is best for the customer's baseboard to go through the oven on a mesh belt. For the second solder paste application, it should also preferably go through the oven on a mesh belt. If, due to special reasons, it cannot go through the oven on a mesh belt, consideration should be given to using a fixture to pass through the oven on a track or placing a flat, high-temperature resistant, flat template to support the PCBA during oven processing to prevent PCB deformation that could cause poor soldering of the module.

8.3.2 Reflow Soldering Operation Instructions

The PCBA reflow oven temperature profile is related to the solder paste used and needs to be adjusted according to the actual solder paste. The data is only suitable for lead-free operations; refer to Figure 8-2 Lead-Free Reflow Soldering Operation Guide.

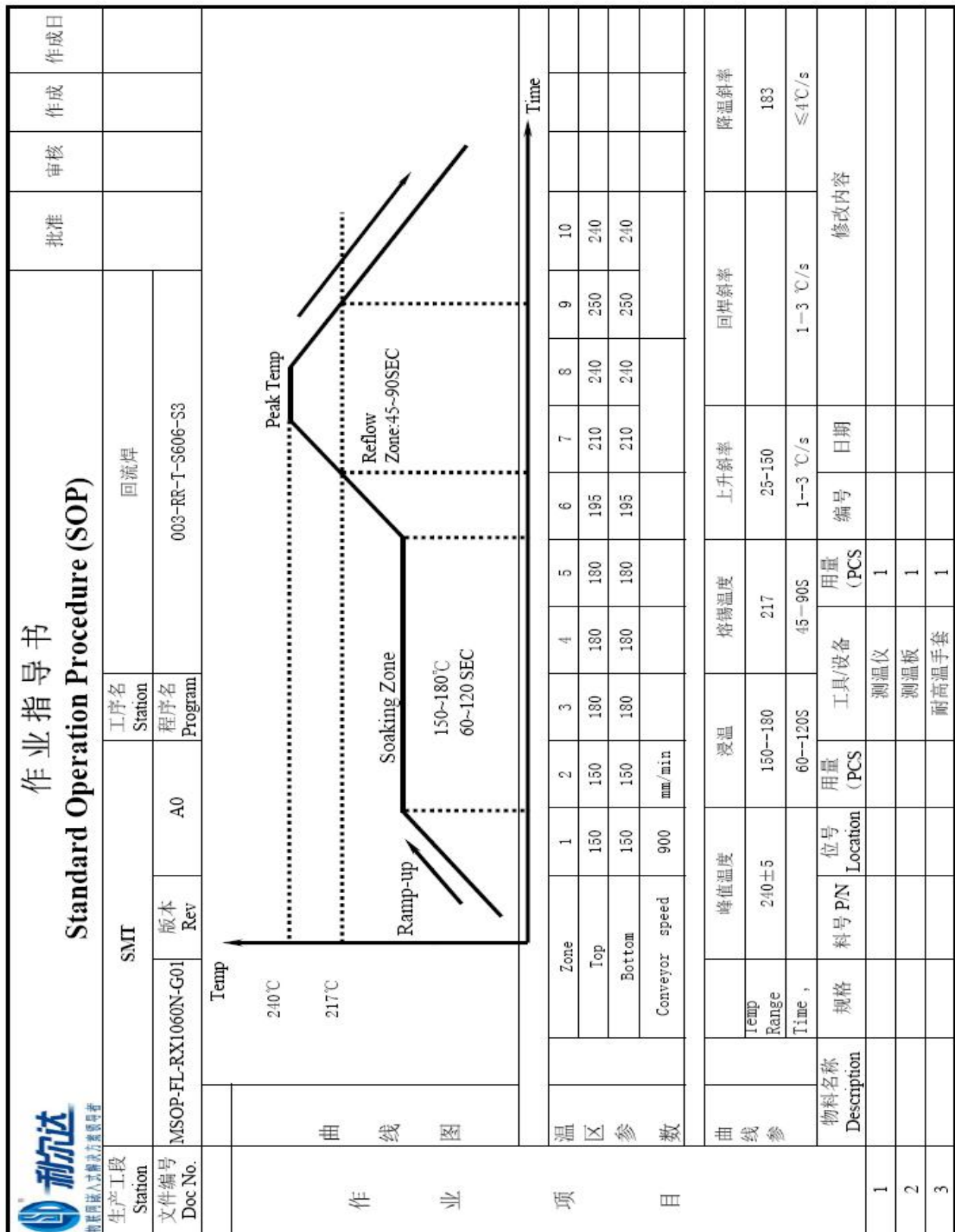


Figure 8-2 Lead-free Reflow Soldering Operation Guide

8.3.3 Production process

During the production welding or other processes that may directly contact the module, no organic solvents (such as alcohol, isopropanol, acetone, trichloroethylene, etc.) shall be used to wipe the module shielding cover; otherwise, it may cause the shielding cover to rust.

If spraying or potting is required for the module, please ensure that the spraying or potting materials used do not chemically react with the module's shield or PCB, and also ensure that the spraying or potting materials do not flow into the interior of the module.

8.3.4 Maintenance

If the module has defects such as cold solder joints or short circuits that require repair, please follow the parameters below:

Lead-free process: Soldering iron temperature $380\pm 10^{\circ}\text{C}$, soldering iron contact time $\leq 5\text{S}$;

Leaded process: Soldering iron temperature $350\pm 10^{\circ}\text{C}$, soldering iron contact time ≤ 5 seconds;

It is not recommended to use a heat gun on the module to avoid affecting its performance.

9 Related Documents and Terminology Abbreviations

9.1 Relevant documents

The following related documents provide the names of the documents, with the version based on the latest release.

Table 9-1 Related Documents

Serial Number	Document Name	Annotation
[1]	Lierda NT26-FCN C Series (Wide Voltage Version) Hardware Reference Design Manual	

9.2 Terminology Abbreviation

Table 9-2 Terminology Abbreviations

Abbreviation	Full English name	Full Chinese name
3GPP	3rd Generation Partnership Project	Third Generation Partner Program
ADC	Analog-to-Digital Converter	Analog-to-digital conversion
ANT	Antenna	Antenna
DAC	Digital -to- Analog Converter	Analog-to-digital conversion
DBG	Debug	Debugging
DC-DC	Direct Current - Direct Current	DC converter
DCXO	Digitally Controlled Crystal Oscillator	Digital Control Crystal Oscillator
DRX	Discontinuous Reception	Non-continuous reception
DTE	Data Terminal Equipment	Data terminal equipment
ECL	Equivalent Class Level	Network Coverage Level
ESD	Electro-Static discharge	Electrostatic discharge

EOS	Electrical Overtress	Electrical Overstress (Surge)
ESR	Equivalent Series Resistance	Equivalent Series Resistance
EVK	Evaluation Kit	Evaluation Toolkit
H-FDD	Half Frequency Division Duplexing	Frequency Division Half Duplex
FOTA	Firmware Over-The-Air	Remote firmware upgrade
GPIO	General-purpose input/output	General Input and Output
I/O	Input/Output	Input and output interface
I _{max}	Maximum Load Current	Maximum current
I _{norm}	Normal Current	Normal (typical) current
bps	Bits Per Second	Rate unit
LCC	Leadless Chip Carriers	Leadless Chip Carrier Packaging
LDO	Low Dropout Regulator	Low dropout linear regulator
LGA	Land Grid Array	Grid array packaging
LwM2M	Lightweight Machine-To-Machine	Lightweight M2M Protocol
MCU	Mirco Controller Unit	Microcontroller Unit
MSL	Moisture Senticity levels	Moisture Sensitivity Level
NB-IoT	Narrow Band Internet of Things	Narrowband Internet of Things (NB-IoT)
PCB	Printed Circuit Board	Printed Circuit Board
PCBA	Printed Circuit Board Assembly	Printed Circuit Board Assembly
PMU	Power Management Unit	Power Management Unit
PSM	Power Saving Mode	Energy-saving mode
RF	Radio Frequency	Radio Frequency
RoHS	Restriction of Hazardous Substances	Restriction of

		Hazardous Substances
RXD	Receive external Data	Receive
TAU	Tracking Area Update	Tracking area update
TCP/IP	Transmission Control Protocol/Internet Protocol	Transmission Control Protocol/Internet Protocol
TVS	Transient Voltage Suppressor	Transient suppression diode
TXD	Transmit external Data	Send
UART	Universal Asynchronous Receiver & Transmitter	Universal Asynchronous Receiver and Transmitter
UDP/IP	User Datagram Protocol - Internet Protocol	User Datagram Protocol
USIM	Universal Subscriber Identification Module	General User Identification Module
VSWR	Voltage Standing Wave Ratio	Voltage Standing Wave Ratio
Vmax	Maximum Voltage Value	Maximum voltage
Vnorm	Normal Voltage Value	Normal (typical) voltage
Vmin	Minimum Voltage Value	Minimum voltage
VIHmax	Maximum Input High Level Voltage Value	Maximum input high level
VIHmin	Minimum Input High Level Voltage Value	Minimum input high level
VILmax	Maximum Input Low Level Voltage Value	Maximum input low level
VILmin	Minimum Input Low Level Voltage Value	Minimum input low level
VImax	Absolute Maximum Input Voltage Value	Maximum input level
VImin	Absolute Minimum Input Voltage Value	Minimum input level
VOHmax	Maximum Output High Level Voltage Value	Maximum output high level
VOHmin	Minimum Output High Level Voltage Value	Minimum output high level

VOLmax	Maximum Output Low Level Voltage Value	Maximum output low level
VOLmin	Minimum Output Low Level Voltage Value	Minimum output low level